

Radar HRRP Target Recognition based on K-SVD Algorithm

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Abstract

We develop a K-SVD algorithm based radar HRRP target recognition method, which can adaptively select the sparse decomposition coefficient based on test noise environment. Experimental results on measured data show our method obtains a better performance than PCA based method.

Keywords: Radar automatic target recognition (RATR); High-resolution range profile (HRRP); Sparse representation; Dictionary Learning; K-SVD algorithm.

1. Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight. It contains the target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition community [1]-[2], [10].

Generally, we should sample a HRRP signal with a higher rate than the effective dimension of the process [3]. However, the additional dimension will make the calculation and storage increase, even cause the curse of dimensionality. Therefore, we need to reduce the dimension of the sampled data to the effective dimension of the underlying process.

The traditional dimensionality reduction algorithms based HRRP target recognition methods can be divided into two approaches: one is based on reconstruction model like Principal Component Analysis (PCA) [1]-[2], Factor Analysis (FA) [2], etc.; another approach is based on the discriminative model, such as Linear Discriminant Analysis (LDA) [4]. All of these methods have an underlying assumption: the number of all observable processes is smaller than the maximal dimension. However, there is no reason to believe that assumption. We hope the representation of HRRP data could be overcomplete, i.e., that the number of physical processes used for HRRP data description can be greater than the signal dimension. Dictionary learning for sparse signal approximation is such a new method for dimensionality reduction. This method has several advantages, including the fact that it encourages a simple model which can avoid over-fitting. In addition, it has physical/biological meaning for model interpretation.

A generalization of the K-means algorithm for dictionary learning, called the K-SVD algorithm, which has been proposed by M. Aharon et al. in 2006 [6], has shown excellent performance in image denoising and inpainting.

Nevertheless, most dictionary learning algorithms for sparse signal approximation are mainly used in natural image processing recently. In our paper, we develop a radar HRRP target recognition based on K-SVD algorithm.

Experimental comparisons with some other existing radar HRRP target recognition algorithms based on reconstruction model, such as PCA-based minimum reconstruction error approximation [1]-[2], demonstrate the better recognition rate and the robustness to noise of our method on the measured radar HRRP data set. The proposed approach adopts an adaptive threshold to determine the sparse decomposition coefficients in order to improve the recognition performance. Conventional radar HRRP recognition algorithms have to divide HRRP samples into different angular domain frames, while the proposed approach can learn the overcomplete dictionary by classes, hence the storage is reduced.

2. Review of Dictionary Learning

2.1. Implementation of dictionary learning

Consider a linear model $\mathbf{D}\mathbf{w} = \mathbf{x}$, where $\mathbf{x} \in \mathbb{R}^P$ and $\mathbf{D} \in \mathbb{R}^{P \times K}$ ($\mathbf{D}$ is a full-rank matrix as $P < K$). An infinite number of solutions are available for this model, but the solution $\mathbf{w}$ with the fewest number of nonzero coefficients is certainly an appealing representation.

There are two stages for the implementation of dictionary learning.

2.1.1. Sparse coding

With the given dictionary $\mathbf{D} \in \mathbb{R}^{P \times K}$ and signal $\mathbf{x} \in \mathbb{R}^P$, we can get the sparse representation $\mathbf{w} \in \mathbb{R}^K$ by solving

$$\min_{\mathbf{w}} \|\mathbf{w}\|_0 \quad s.t. \quad \|\mathbf{x} - \mathbf{D}\mathbf{w}\|_2^2 \leq C\sigma^2 \quad (1)$$

$$\text{or} \quad \min_{\mathbf{w}} \|\mathbf{x} - \mathbf{D}\mathbf{w}\|_2^2 \quad s.t. \quad \|\mathbf{w}\|_0 \leq T \quad (2)$$

where $\|\mathbf{w}\|_0$ represents the l_0 norm, counting the number of non-zero elements in $\mathbf{w}$. C is a constant, σ^2 and T represent the noise variance and sparsity level, respectively. However, the exact solution of (1) or (2) requires searching over all combinations of non-zero components, which is a NP-hard problem. There are two ways to get an approximate solution. One approach is to relax the l_0 constraint to a l_1 constraint, yielding a convex problem, which can be solved by basis pursuit (BP) [7]. Another way is using greedy algorithms, such as orthogonal matching pursuit (OMP), matching pursuit (MP) [9].

2.1.2. Dictionary learning

Designing dictionaries to better fit the linear model $\mathbf{D}\mathbf{w} = \mathbf{x}$ can be done by either selecting one from a prespecified set or adapting the dictionary to a set of training signals. Many prespecified dictionaries, such as discrete fourier transform (DFT), discrete cosine transform (DCT), wavelet and their overcomplete versions, are appealing due to their simple computations. But using dictionaries adapted to the data, such as K-SVD and method of optimal directions (MOD), might better fit the signal of interest, leading to improved performance over a prespecified dictionary.

2.2. K-SVD algorithm

K-SVD is an iterative algorithm for designing overcomplete dictionaries for sparse representations. It alternates between the sparse coding of the examples based on the current dictionary and a process of updating the dictionary atoms to better fit the data until reaching the stopping rule. SVD is used in K-SVD to implement the closest rank-1 approximation to its representation error as

$$\tilde{\mathbf{d}}_k \tilde{\mathbf{w}}_{k:} \approx \mathbf{X} - \sum_{j \neq k} \mathbf{d}_j \mathbf{w}_{j:} \quad (3)$$

where $\tilde{\mathbf{d}}_k$ and $\tilde{\mathbf{w}}_{k:}$ represent the updated k th atom of the dictionary and k th row of the sparse representation, respectively.

3. K-SVD Algorithm based Radar HRRP Recognition

3.1. The preprocessing of the measured HRRP data

In the statistical target recognition based on HRRP, a challenging task is how to deal with the time-shift and amplitude-scale sensitivity of HRRP.

3.1.1. Time-Shift Sensitivity

In order to decrease the computation complexity, an HRRP is only a part of received radar echo extracted by a range window, in which target signal is included. Thus the position of the target signal in an HRRP can vary with the measurement. There are many useful feature extraction methods, such as the spectrum amplitude of real HRRPs, power spectrum and etc., to deal with the time-shift sensitivity of HRRP. With the given HRRP $x(t)$, the Fourier transform of the $x(t - \tau)$ can be represented as:

$$X'(w) = X(w) \exp(-jw\tau) \quad (4)$$

$$|X'(w)| = |X(w)| \quad (5)$$

Obviously, the spectrum amplitude of real HRRPs is a feature extraction method invariant with the initial phases for complex HRRP samples. So we take the spectrum amplitude of real HRRPs as the feature in our method. Because the spectrum amplitude has a symmetrical structure, half of the spectrum amplitude sequence can be used as the input signal.

3.1.2. Amplitude-Scale Sensitivity

The amplitude-scale sensitivity of HRRP comes from the fact that the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar

receiver gain, radar system losses and so on. Thus HRRPs measured by different radars or under different conditions will have different amplitude-scales. Here we take amplitude normalization to deal with the amplitude-scale sensitivity of HRRPs.

3.2. Recognition Method

The procedures of the K-SVD algorithm based radar HRRP recognition can be expressed as:

Table 1.

K-SVD Algorithm based Radar HRRP Recognition Method

■ Dictionary Learning Stage:

1. Preprocess: Training samples $\mathbf{X}_l \in \mathbb{R}^{P \times N_l}$ ($l = 1, 2, \dots, L$) are defined as the spectrum amplitude of real HRRPs with amplitude normalization.
2. Initialization: Set the dictionary matrix $\mathbf{D}^{(0)} \in \mathbb{R}^{P \times K}$ with l_2 normalized columns. Set $i = 1$.
3. Repeat until reaching the stopping rule:

◆ Sparse coding:

With a stopping criterion defined by the noise variance or sparsity level, OMP is used to compute $\mathbf{W}^{(i)}$.

$$\min_{\mathbf{W}} \|\mathbf{X}_l - \mathbf{D}_l^{(i)} \mathbf{W}_l^{(i)}\|_F^2 \quad s.t. \quad \forall n, \|\mathbf{w}_{ln}^{(i)}\|_0 \leq T \quad (6)$$

◆ Dictionary Update:

$$\begin{aligned} \|\mathbf{X}_l - \mathbf{D}_l^{(i)} \mathbf{W}_l^{(i)}\|_F^2 &= \left\| \mathbf{X}_l - \sum_{j=1}^K \mathbf{d}_{lj}^{(i)} \mathbf{w}_{lj:}^{(i)} \right\|_F^2 \\ &= \left\| \left(\mathbf{X}_l - \sum_{j \neq k} \mathbf{d}_{lj}^{(i)} \mathbf{w}_{lj:}^{(i)} \right) - \mathbf{d}_{lk}^{(i)} \mathbf{w}_{lk:}^{(i)} \right\|_F^2 \\ &= \left\| \mathbf{E}_{lk} - \mathbf{d}_{lk}^{(i)} \mathbf{w}_{lk:}^{(i)} \right\|_F^2 \end{aligned} \quad (7)$$

Apply SVD decomposition to update the atoms.

◆ Set $i = i + 1$.

■ HRRP target Recognition stage:

1. Preprocess: Normalize the spectrum amplitude of a test HRRP sample denotes as $\mathbf{y}$.
2. Coefficient selection: Estimate the SNR of $\mathbf{y}$, then compare the SNR with adaptive threshold to select the sparse decomposition coefficient T^* .
3. Sparse coding: Compute $\boldsymbol{\alpha}_l$ by OMP with the adaptive sparse decomposition coefficient T^* .
4. Recognition: Compute the reconstruction error e_l by formula (8), and then determine the class label l^* by

formula (9).

$$e_l = \|\mathbf{y} - \mathbf{D}_l^* \mathbf{a}_l\|_2 \quad (8)$$

$$l^* = \arg \min_l e_l \quad (l = 1, 2, \dots, L) \quad (9)$$

In evaluating the computational complexity of the algorithm, we consider the HRRP target recognition stage only, the stage requires $\mathcal{O}((T^* + P) \cdot K)$ operations, where P is the dimension of sample, K is the number of atoms in dictionary and T^* is the adaptive sparse decomposition coefficient.

4. Experimental Results

We carried out several experiments on HRRP data, trying to show the practicality of the proposed algorithm. The parameters of the targets and radar are shown in Table 2, and the projections of target trajectories onto the ground plane are shown in Figure 1, from which the aspect angle of an airplane can be estimated according to its relative position to radar. The original training set in our experiments is the same as that used in [1]–[2], with 35 aspect-frames for Yak-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each frame has 1024 HRRP samples. In addition, the measured HRRP is a 256-dimensional vector.

Table 2.

Parameters of planes and radar in the ISAR experiment

radar parameters	Center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	4.57

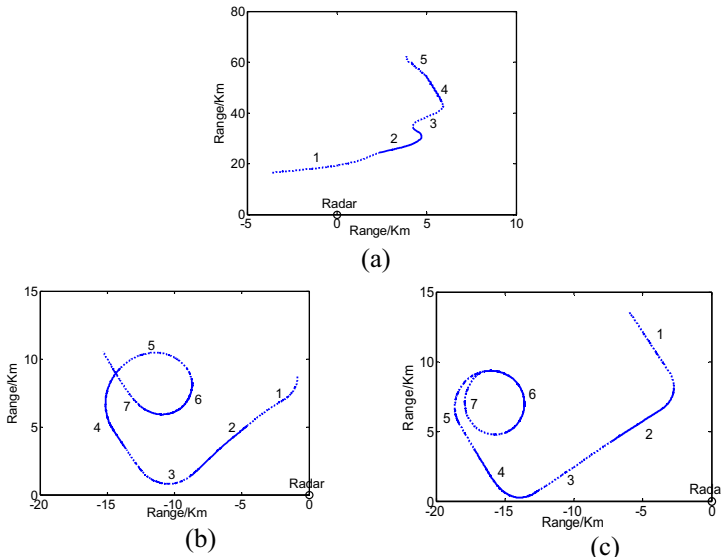


Fig.1. Projections of target trajectories onto the ground plane: (a) Yark-42; (b) An-26; (c) Cessna Citation S/II.

White Gaussian noise with varying signal-to-noise ratio (SNR) is added to the measured HRRPs. According to the introduction in [1]–[2], the SNR in the measured data is very high. For aircraft-like targets, the noise in the inphase and

quadrature echoes of targets can be assumed to be white Gaussian noise. In order to evaluate the effect of noises on the proposed model and other existing statistical models, we add simulated complex Gaussian white noises to the raw test HRRP data. The SNR is defined as:

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\bar{P}_s}{P_{\text{Noise}}} \right) = 10 \times \log_{10} \left(\frac{\sum_{n=1}^N P_{sn}}{N \times P_{\text{Noise}}} \right) \quad (10)$$

where $\bar{P}_s$ denotes the average power of original HRRP, P_{sn} denotes the power of the original echo per range cell, N denotes the number of range cells, and P_{Noise} denotes the power of noise. Thus $10^{\text{SNR}/10}$ equals to the rate of the average power of original HRRP to the power of the noise.

We applied the K-SVD, training a class dictionary of size 128×768 . The coefficients were computed using OMP with a fixed number of coefficients, where the maximal number of coefficients is three. The maximum number of the iteration is 10. The proposed approach can learn the overcomplete dictionary by classes and needs not to divide HRRP samples into different angular domain frames, hence the storage is reduced.

Fig.2 shows that different selections of the sparse decomposition coefficient for OMP sparse decomposition can get distinct recognition rate in our experiment.

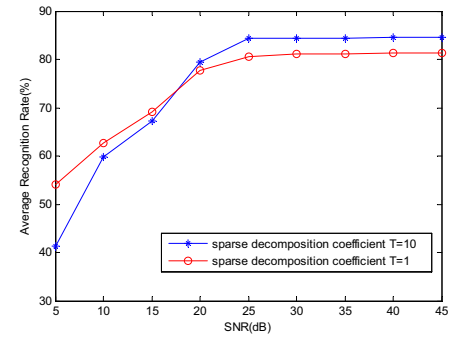


Fig.2. Variation of the recognition performance with SNR, via the K-SVD algorithm with fixed sparse decomposition coefficient $T=1$ and $T=10$.

When the test sample is a low-SNR signal, we should use only a few dictionary atoms to represent the signal in order to reduce the influence of noise. When the test sample is a high-SNR signal, we should use more dictionary atoms to represent the signal, and in this case it can describe the signal more accurately. The average recognition rate of the proposed algorithm shown in Fig.3 and 4 supports our view.

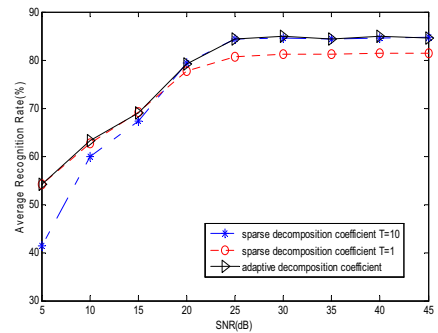


Fig.3. Variation of the recognition performance with SNR, via the K-SVD algorithm with adaptive sparse decomposition coefficient and fixed decomposition coefficient ($T=1$ and 10).

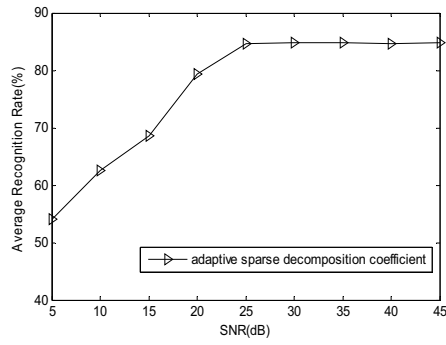


Fig.4. The average recognition rate of the proposed K-SVD based radar HRRP target recognition algorithm with adaptive sparse decomposition coefficient.

Fig.5 shows that the proposed K-SVD algorithm based radar HRRP recognition method with adaptive sparse decomposition coefficient has a better recognition rate than the PCA-based minimum reconstruction error approximation [1]-[2] method. Furthermore, the proposed method is more robust than the PCA-based minimum reconstruction error approximation method in the low-SNR case.

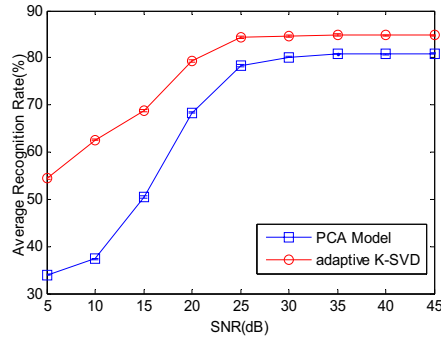


Fig.5. Variation of the recognition performance with SNR, via the proposed K-SVD based radar HRRP target recognition algorithm and PCA-based minimum reconstruction error approximation [1]-[2]. The recognition rates are averaged over twenty experiments, and the error bars denote the corresponding standard deviations.

5. Conclusion

The K-SVD algorithm based radar HRRP recognition method proposed in this paper utilizes the dictionary learning for sparse signal approximation to radar HRRP target recognition. Experimental results indicate that the recognition rate and the robustness to noise of our method are likely to be statistically superior to PCA-based minimum reconstruction error approximation for measured HRRP data sets.

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